Vorlesung 26

Kovariante und Kontravariante Vektoren

A_i ← unten

A^i ← oben

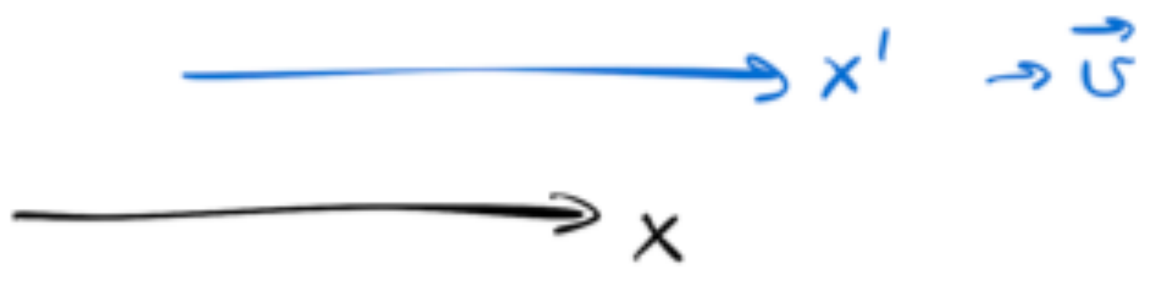
$$\text{Skalarprodukt} = g_{ij} A^i A^j = A_j A^j$$

A_j

ohne g_{ij} !

$$\{A_j\} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} A^0 \\ A^1 \\ A^2 \\ A^3 \end{pmatrix} = (A^0, -A^1, -A^2, -A^3)$$

Lorentz-Tr.



$$A'^0 = \frac{A^0 - \frac{v}{c}A^1}{\sqrt{\dots}}$$

$$A'^1 = \frac{-\frac{v}{c}A^0 + A^1}{\sqrt{\dots}}$$

kontra ....

$A^0,\ A'^0 \to A_0,\ A'_0$

$A^{1,2,3},\ A'^{1,2,3} \to -A_{1,2,3},\ -A'_{1,2,3}$

$$A'_0 = \frac{A_0 + \frac{v}{c}A_1}{\sqrt{\dots}}$$

$$A'_1 = \frac{+\frac{v}{c}A_0 + A_1}{\sqrt{\dots}}$$

kovariant

$(c\,dt, dx, dy, dz) \leftarrow$ kontrav.

$\left(\frac{\partial}{c\,\partial t};\ \frac{\partial}{\partial x};\ \frac{\partial}{\partial y};\ \frac{\partial}{\partial z}\right) \leftarrow$ kovariant

$$\varphi(ct, x, y, z) \to \varphi(ct', x', y', z')$$

$$\frac{\partial\varphi}{c\,\partial t'} = \frac{\partial\varphi}{c\,\partial t}\frac{c\,\partial t}{c\,\partial t'} + \frac{\partial\varphi}{\partial x}\frac{\partial x}{c\,\partial t'}$$

$$\frac{\partial\varphi}{\partial x'} = \frac{\partial\varphi}{c\,\partial t}\frac{c\,\partial t}{\partial x'} + \frac{\partial\varphi}{\partial x}\frac{\partial x}{c\,\partial x'}$$

koeffizienten der inversen Lorentz-Tr.

$t(t', x')$
$x(t', x')$ $\Big|\leftarrow \vec{v} \to -\vec{v}$

Kontinuitätsgl.: $$\frac{\partial \varrho}{\partial t} + (\vec{\nabla} \cdot \vec{J}) = 0$$

$$\frac{c\,\partial \varrho}{c\,\partial t} + \frac{\partial J_x}{\partial x} + \frac{\partial J_y}{\partial y} + \frac{\partial J_z}{\partial z} = 0$$

$$\underbrace{\left(\frac{\partial}{c\,\partial t} ; \frac{\partial}{\partial x} ; \frac{\partial}{\partial y} ; \frac{\partial}{\partial z} \right)}_{\text{kovarianter 4-Vektor}} \cdot \begin{pmatrix} c\varrho \\ J_x \\ J_y \\ J_z \end{pmatrix} = \underset{\substack{\uparrow \\ \text{4-Skalar}}}{0}$$

$$\Downarrow$$

$\begin{pmatrix} c\varrho \\ \vec{J} \end{pmatrix}$ ist kontravarianter 4-Vektor

Ähnlich $\longrightarrow$ Lorenz-Eichung: $$\frac{\partial \Phi}{\partial t} + \vec{\nabla} \cdot \vec{A} = 0$$

$$\Downarrow$$

$\begin{pmatrix} \Phi \\ \vec{A} \end{pmatrix}$ ist kontravarianter 4-Vektor

Relativistische $\underbrace{\text{Wirkung}}_{S}$ und $\underbrace{\text{Lagrange-Funktion}}_{L}$

Nicht relativistisch: $S = \int_{t_1}^{t_2} L(\dot{q}_i, q_i)\, dt$

$$L = E_{kin} - E_{pot}$$

Variationsverlauf $\rightarrow$ Euler-Lagrange Gln.: $\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) = \frac{\partial L}{\partial q_i}$

Relativistische Beschreibung $\Leftrightarrow$ relativistisch invariante Gesetze

$\Downarrow$

S muss 4-Skalar sein!

Freies Teilchen: Die Eigenzeit ist das "einzige" verfügbare 4-Skalar!

$$\Rightarrow \quad S = \underbrace{-mc^2}\int_{\{x_1^i\}}^{\{x_2^i\}} d\tau = \int \underbrace{(-mc^2)\sqrt{1-\frac{v^2}{c^2}}}_{L}\, dt$$

abgestimmt von dem nicht rel. Grenzfall

$$\left.\begin{aligned} \vec{q} &= \vec{r} \\ \dot{\vec{q}} &= \vec{v} \end{aligned}\right] \Rightarrow \frac{\partial L}{\partial \dot{\vec{q}}} = \frac{\partial L}{\partial \vec{v}} = \frac{(\cancel{-}mc^2)(\cancel{-}\cancel{2}\frac{\vec{v}}{c^2})}{\cancel{2}\sqrt{1-\frac{v^2}{c^2}}} = \boxed{\frac{m\vec{v}}{\sqrt{1-\frac{v^2}{c^2}}} = \vec{p}} \quad \Bigg| \quad \frac{\partial L}{\partial \vec{q}} = \frac{\partial L}{\partial \vec{r}} = \vec{0}$$

Euler-Lagr.: $\quad \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\vec{q}}}\right) = \frac{\partial L}{\partial \vec{q}} \Rightarrow \boxed{\frac{d\vec{p}}{dt} = \vec{0}}$

Energie: $\quad \boxed{E = \vec{p}\cdot\vec{v} - L} = \frac{mv^2}{\sqrt{1-\frac{v^2}{c^2}}} + \frac{mc^2\sqrt{1-\frac{v^2}{c^2}}\,\cancel{\sqrt{\cdots}}}{\cancel{\sqrt{\cdots}}} = \frac{\cancel{mv^2} + mc^2\left(1-\cancel{\frac{v^2}{c^2}}\right)}{\sqrt{\cdots}} =$

$$= \boxed{\frac{mc^2}{\sqrt{1-\frac{v^2}{c^2}}}}$$

Teilchen + EM Feld

Ladung

$$S = \int_1^2 (-mc^2)\, d\tau + (4\text{-Vekt})_i\, dx^i = \int_1^2 (-mc^2)\, d\tau - \frac{q}{c} A_i\, dx^i =$$

$$\{A_i\} = \begin{pmatrix} \varphi \\ -\vec{A} \end{pmatrix}$$

$$= \int_1^2 \left[-mc^2 \sqrt{1-\frac{v^2}{c^2}}\, dt - \frac{q}{c} \left(\varphi\, c\, dt - \vec{A} \cdot \frac{d\vec{r}}{dt} dt \right) \right. =$$

$$= \int_{t_1}^{t_2} \left[-mc^2 \sqrt{1-\frac{v^2}{c^2}} + \frac{q}{c} \vec{A} \cdot \vec{v} - q\varphi \right] dt$$

Verallg. Impuls: $\boxed{\vec{P} = \frac{\partial L}{\partial \vec{v}} = \frac{m\vec{v}}{\sqrt{1-\frac{v^2}{c^2}}} + \frac{q}{c}\vec{A}} = \vec{p} + \frac{q}{c}\vec{A}$

$$\vec{b}\,(\vec{c}\cdot\vec{a})$$

$$\frac{\partial L}{\partial \vec{r}} = \frac{q}{c} \vec{\nabla}(\vec{A}\cdot\vec{v}) - q\vec{\nabla}\varphi$$

$$\vec{a} \times (\vec{b} \times \vec{c}) = \vec{b}(\vec{a}\cdot\vec{c}) - (\vec{a}\cdot\vec{b})\vec{c} \Rightarrow \vec{\nabla}(\vec{A}\cdot\vec{v}) = (\vec{v}\cdot\vec{\nabla})\vec{A} + \vec{v}\times(\vec{\nabla}\times\vec{A})$$

Eul. – Lagr.

$$\frac{d}{dt}\frac{\partial L}{\partial \vec{v}} = \frac{\partial L}{\partial \vec{r}} \Rightarrow \frac{d}{dt}\Big(\underbrace{\vec{p}}\ + \underbrace{\frac{q}{c}\vec{A}}\Big) = \cancel{\frac{q}{c}(\vec{v}\cdot\vec{\nabla})\vec{A}} + \frac{q}{c}\vec{v}\times(\vec{\nabla}\times\vec{A}) - q\vec{\nabla}\varphi$$

$$\frac{q}{c}\frac{d\vec{A}(t,\vec{r}(t))}{dt} = \frac{q}{c}\,\frac{\frac{\partial A}{\partial t}dt + \frac{\partial \vec{A}}{\partial r_i}dr_i}{dt} = \frac{q}{c}\frac{\partial \vec{A}}{\partial t} + \cancel{\frac{q}{c}\Big(\underbrace{\frac{d\vec{r}}{dt}}_{\vec{v}}\cdot\vec{\nabla}\Big)\vec{A}}$$

$$\frac{d\vec{p}}{dt} = \underbrace{-\frac{q}{c}\frac{\partial \vec{A}}{\partial t} - q\vec{\nabla}\varphi}_{} + \frac{q}{c}\vec{v}\times\underbrace{(\vec{\nabla}\times\vec{A})}_{\vec{B}}$$

$$\vec{E} = -\frac{\partial \vec{A}}{c\,\partial t} - \vec{\nabla}\varphi$$

$\rightarrow$ Gauß. Einheiten

$$\boxed{\frac{d\vec{p}}{dt} = q\vec{E} + \frac{q}{c}\vec{v}\times\vec{B}}$$

$$\uparrow$$

$$\vec{p} = \frac{m\vec{v}}{\sqrt{1-\frac{v^2}{c^2}}}$$

Zum Weiterlesen:

L.&L. § 6, 7, 8, 9, 11, 15, 16, 17

Jackson 11.5, 11.6, 11.7, 12.1